

Tribhuvan University

Faculty of Humanities and Social Science

[Your Campus / College Name]

Department of Computer Applications

A PROJECT REPORT

ON

FinSight – Smart Personal Finance & Budget Tracker

Submitted to

Department of Computer Applications

[College Name], [Location]

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by:

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BCA 6th Semester, III Year

Under the supervision of

[Supervisor Name]

Tribhuvan University

Faculty of Humanities and Social Science

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SUPERVISOR’S RECOMMENDATION

I hereby recommend that this project prepared under my supervision by **[Student Name 1]** and **[Student Name 2]** entitled “**FinSight – Smart Personal Finance & Budget Tracker**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

Name of the Supervisor:

[Supervisor Name]

Signature of the Supervisor

BCA 6th Semester, III Year

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LETTER OF APPROVAL

This is to certify that this project prepared by **[Student Name 1]** and **[Student Name 2]** entitled **“FinSight”** in partial fulfillment of the requirements for the degree of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

SUPERVISOR [Supervisor Name] Contact no: [Phone] Department of BCA [College Name]	HOD / CO-ORDINATOR [HOD Name] Contact no: [Phone] Department of BCA [College Name]
INTERNAL EXAMINER Name: [Name] Date: _____	EXTERNAL EXAMINER Name: [Name] Date: _____

ABSTRACT

FinSight is a modern personal finance and budget tracking platform designed to enable intelligent expense management, personalized spending insights, and proactive financial control. It features support for secure registration of users, logging of income and expense transactions with categories and notes, dynamic account balance calculations, and interactive engagement with financial data through budgets, savings goals, and real-time alerts. With the help of the Z-Score Anomaly Detection algorithm, the app ensures unusual expenses are dynamically flagged by comparing each transaction against the user's own historical spending in the same category. A Multinomial Naive Bayes categorizer integrated with Levenshtein distance typo matching suggests categories in real-time as users type descriptions, while 1D K-Means spending clustering categorizes expenses into micro, regular, and major purchasing tiers. Additionally, a Budget Pace Prediction algorithm projects end-of-month totals and warns users before limits are exceeded.

FinSight comes with a responsive and minimalist user interface built with modern React components and TailwindCSS, which is available on desktop and mobile browsers. FastAPI is utilized for solid backend operations and React.js for the frontend to provide real-time interaction and easy navigation. Functionality includes transaction CRUD with soft deletion, budget monitoring with rollover, savings goal tracking with goal contribution transfers, recurring transaction detection, CSV import, and an analytics dashboard for tracking metrics such as net worth and category spending breakdowns. Data privacy is maintained through complete isolation of user data with JWT-based authentication. Overall, FinSight provides an end-to-end digital space for personal finance tracking, discovery of spending patterns, and proactive decision support — bridging the gap between simple spreadsheets and intelligent financial assistants.

Keywords: Personal Finance, Budget Tracking, Anomaly Detection, Z-Score, Naive Bayes, Levenshtein Distance, 1D K-Means Clustering, Budget Pace Prediction, FastAPI, React.js, PostgreSQL, JWT Authentication.

ACKNOWLEDGEMENT

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We extend our genuine thankfulness to our project supervisor **[Supervisor Name]** for his/her valuable and great support and guidance in all the happenings regarding to the project and also we must express our thanks to **[HOD Name]**, HOD for his/her encouragement and guidance throughout the project.

We are indebted to the department of computer application for providing support to add on our venture. Last, but not the least, we would like to thank our teachers and colleagues who have been knowingly or unknowingly the part of this project and let support and views during the entire development time.

Your sincerely,

[Student Name 1] ([Reg No])

[Student Name 2] ([Reg No])

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LIST OF ABBREVIATIONS

API	Application Programming Interface
BCA	Bachelor of Computer Application
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
CSV	Comma-Separated Values
FK	Foreign Key
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
JWT	JSON Web Token
JSON	JavaScript Object Notation
K-Means	K-Means Clustering Algorithm
NB	Multinomial Naive Bayes
NPR	Nepalese Rupee
ORM	Object-Relational Mapping
PDF	Portable Document Format
REST	Representational State Transfer
SQL	Structured Query Language
TU	Tribhuvan University
UI	User Interface
UX	User Experience
UML	Unified Modeling Language
UUID	Universally Unique Identifier

CHAPTER 1: PROJECT INTRODUCTION

1.1 Introduction

With the age of digital consumption and personal financial complexity, developing significant control over money and sharing valuable insights requires an updated approach. This project is concerned with designing an accessible and comprehensive personal finance platform “FinSight” as a valuable asset that will transform the way digital money is tracked, analysed, and managed.

FinSight is a modern personal finance platform that offers the facility of an all-inclusive transaction and budget management system. Only registered members are allowed to login to the system and new members can be registered in the application through secure authentication processes. The system is developed in such a way that it contains a user module focused on personal financial intelligence. The user module possesses an advanced platform with individualized spending streams, anomaly alerts, real-time category suggestions using Naive Bayes and typo matching, 1D K-Means spending clustering, and budget pace prediction to facilitate users in understanding their own financial behavior easily and rapidly. If a user inputs incorrect information, the system presents appropriate error messages and validation hints.

Personal finance management and data-driven decision making are very important aspects of modern digital life, as people increasingly rely on digital tools for expense tracking, budget discipline, and goal achievement. Statistical analysis of spending patterns has revolutionized personal finance by blending traditional bookkeeping techniques with contemporary data science tools, making financial awareness more actionable, captivating, and engaging for individuals. This transformation has created new opportunities for users to understand themselves through anomaly detection, category auto-suggestion, and predictive alerts that combine historical data with real-time insight.

1.2 Problem Statements

FinSight platform aims to address the following challenges that are present in existing platforms:

1. Individuals in Nepal and similar contexts do not have access to a specialized platform where they can track income, expenses, budgets, and savings goals with intelligent statistical analysis beyond simple spreadsheets or basic mobile apps.
2. Existing finance applications lack modern features like personalized Z-Score anomaly detection, Naive Bayes real-time category prediction with typo tolerance, 1D K-Means spending clustering, and budget pace prediction that help users find quality insights into their spending behavior.
3. Users have no way to track their financial performance, dynamic account balances, spending anomalies, and goal progress through a dedicated personal dashboard system that combines multiple indicators into a unified interface.

1.3 Objectives

Some of the key objectives of the project are as follows:

1. To create a specialized platform for individuals where they can log income and expense transactions, set monthly budgets, define savings goals, and receive intelligent alerts with complete control over their own financial data.
2. To implement advanced analytical features including Z-Score anomaly detection, Naive Bayes categorizer with Levenshtein typo matching, 1D K-Means spending clustering, and budget pace prediction that help users discover quality insights into their financial behavior.
3. To provide comprehensive personal dashboards where users can track their spending performance, anomaly activity, goal progress, and manage their financial records effectively.

1.4 Scope and Limitations

1.4.1 Scope

The scope of FinSight platform are as follows:

1. Create a comprehensive personal finance platform where users can log, categorise, and analyse income and expense transactions, set budgets, and track savings goals with complete data privacy.
2. Create responsive web application with modern UI/UX design, dark mode support, and professional interface optimized for financial overview and detailed analysis experience.
3. Provide comprehensive analytical features including Z-Score anomaly detection, Naive Bayes auto-categorization with typo distance matching, 1D K-Means spending clustering, budget pace prediction, recurring transaction detection, CSV import, and global search across all financial records.

1.4.2 Limitations

The limitations of FinSight platform are as follows:

1. Direct bank API integration and automatic transaction fetching is not implemented; users must import CSV files or enter data manually.
2. Mobile application (iOS/Android) is not developed, restricting access to web browsers only (responsive layout is provided).
3. Multi-user / family shared accounts and real-time collaborative budgeting features are not available in the current version.

1.5 Development Methodology

Incremental Development Methodology was used while building this system. The Incremental Methodology works well for projects where core functionality can be delivered early and complex analytical modules can be layered on top. There was specific documentation, evolving requirements for algorithms, and well-understood technology in this project. Thus, Incremental Methodology was suitable for this system. The final deliverables were clearly understood so Incremental Methodology of Project Management seemed perfect for the development of this system.



Figure 1.1: Incremental Development Methodology of FinSight

The following approach was taken to design, develop, and implement the above mentioned system:

Requirement Analysis: The first step is to identify the requirements for the FinSight Finance Platform. This involves conducting a thorough analysis of user roles, transaction logging, budget processes, anomaly detection requirements, categorization needs, and analytics dashboard requirements.

System Design: Based on requirement analysis, the system is designed. This includes selecting appropriate technologies such as React for frontend, FastAPI and Python for backend, PostgreSQL for database, and designing pure Python statistical algorithms for anomaly detection, clustering, and categorization.

Implementation & Coding: In this phase, each module (authentication, transaction creation, anomaly detection engine, Naive Bayes categorizer, K-Means clustering, budget pace prediction, CSV import) is developed and integrated.

Testing: Unit testing and system testing are performed to ensure functionality, performance optimization, and clean error handling.

Deployment: Setting up web server configurations and database migration procedures for production execution.

1.6 Report Organization

The chapter wise organization of the report is as following:

Chapter 1: Introduction of the project including problem statement related to personal finance visibility challenges, objectives for developing project, and scope and limitations of the project.

Chapter 2: Background study and literature review covering fundamental concepts related to the project. Study of existing systems to understand current market solutions and identify gaps.

Chapter 3: System analysis and design including requirement analysis for the project. Includes analysis of functional and non-functional requirements. Includes feasibility analysis. System modeling using various UML diagrams and complete details of statistical algorithms used.

Chapter 4: Implementation and testing details covering the development process using tools for frontend and backend, database design and integration, user interface development, and testing strategies.

Chapter 5: Conclusion and future recommendations summarizing the successful development of project, lessons learned, and future work.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Digital personal finance tracking and analytical platforms have become essential tools for individuals, students, and professionals worldwide. These platforms serve as digital spaces where individuals can record transactions, set budgets, monitor goals, and engage with their financial data. Despite significant growth in digital adoption, users lack sophisticated platforms that cater specifically to personalized statistical insights, real-time category suggestion, and anomaly alerts without exposing sensitive bank login credentials to third-party services.

2.2 Literature Review

Academic studies on financial anomaly detection reveal that users require sophisticated tools for transaction management, performance analytics, and proactive alerts. Research indicates that statistical methods such as Z-Score are efficient tools for flagging unusual transactions. Furthermore, text classification using Multinomial Naive Bayes combined with Levenshtein edit distance provides effective, lightweight category suggestions for sparse financial text descriptions. Unsupervised 1D K-Means clustering effectively segments spending into distinct natural tiers (micro, regular, major) without requiring hardcoded monetary boundaries.

2.3 Study of the Existing System

Existing systems like Mint, YNAB, and Goodbudget offer basic expense tracking or envelope budgeting. However, they lack per-user statistical anomaly detection, intelligent real-time category suggestion with fuzzy typo matching, and dynamic budget pace forecasting tailored to local currency (NPR) contexts.

2.4 Proposed System

The proposed FinSight platform addresses these gaps by implementing a web-based platform featuring Z-Score anomaly detection, Multinomial Naive Bayes categorization with Levenshtein distance, 1D K-Means spending clustering, dynamic balance calculations, and budget pace prediction using FastAPI, React, and PostgreSQL.

FinSight incorporates robust authentication measures, secure data handling, and responsive design principles to ensure seamless access across devices. The platform features a clean, distraction-free interface optimized for financial overview and detailed analysis, with modern UI/UX design that encourages both logging and insight activities. The platform will serve as a catalyst for financial literacy and proactive money management in Nepal's digital landscape.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

3.1.1 Requirement Analysis

i. Functional Requirements:

1. **User Registration & Authentication:** Secure user registration and JWT-based authentication.
2. **Transaction Management:** Full CRUD functionality for income, expense, and transfer records with soft deletion.
3. **Auto-Categorization:** Real-time category suggestions via Naive Bayes and Levenshtein typo distance.
4. **Z-Score Anomaly Detection:** Automatically scoring expenses against historical category baselines.
5. **1D K-Means Spending Clustering:** Grouping expenses into micro, regular, and major tiers.
6. **Budget Pace Prediction:** Projecting end-of-month spending from daily burn rate.

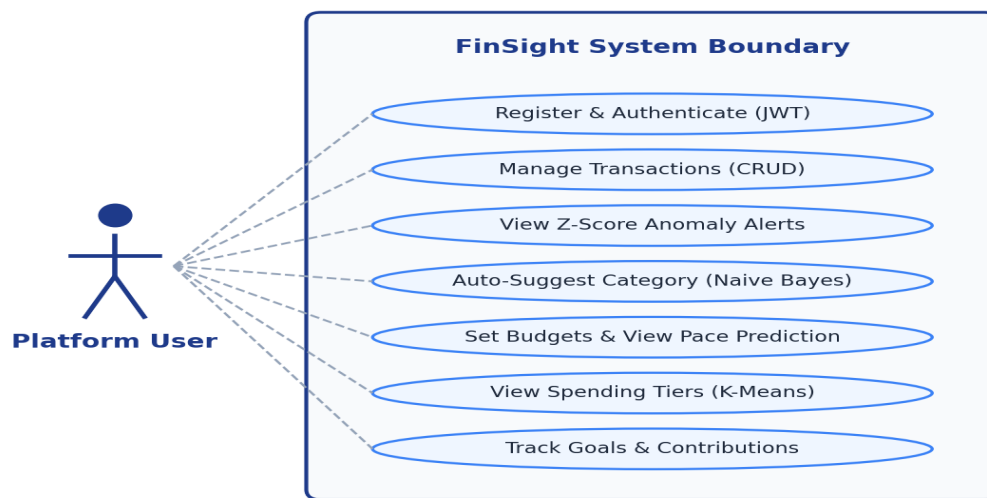


Figure 3.1: Use Case Diagram of FinSight

ii. Non-Functional Requirements

1. **Security:** bcrypt password hashing, JWT tokens signed with HS256.
2. **Usability:** Clean React + TailwindCSS interface.
3. **Reliability:** ACID-compliant PostgreSQL transaction management.
4. **Scalability:** FastAPI asynchronous non-blocking routing.

3.1.2 Feasibility Analysis

Technical, operational, economic, and schedule feasibility were evaluated and confirmed suitable for the academic project constraints.

S.N	Phases	Time Duration
1	Requirement Analysis	1–2 weeks
2	System Design	3–4 weeks
3	Implementation (Increments 1–4)	6–8 weeks
4	Testing	1–2 weeks
5	Documentation & Deployment	1 week

Table 3.1: Project Schedule for FinSight

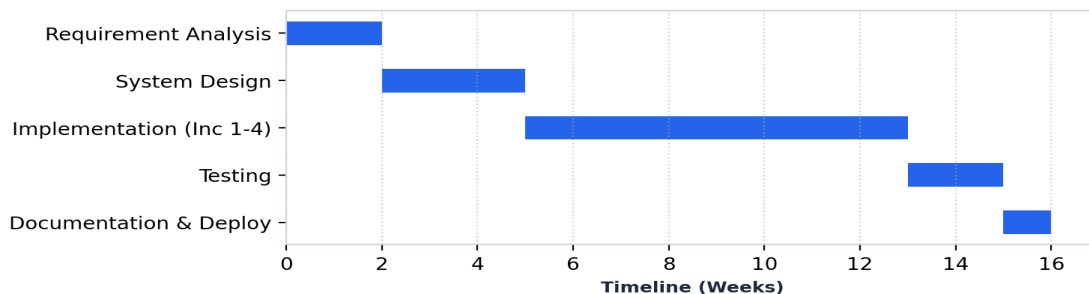


Figure 3.2: Gantt Chart of FinSight

3.1.3 Object Modelling & Class Diagram

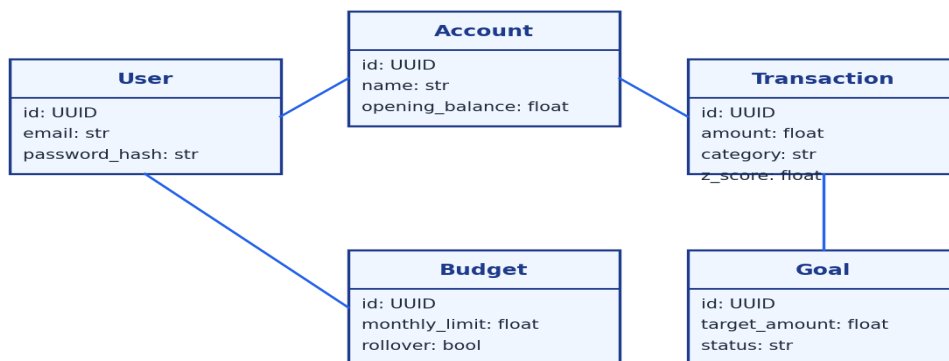


Figure 3.3: Class Diagram of FinSight

3.1.4 Dynamic Modelling & Process Modeling

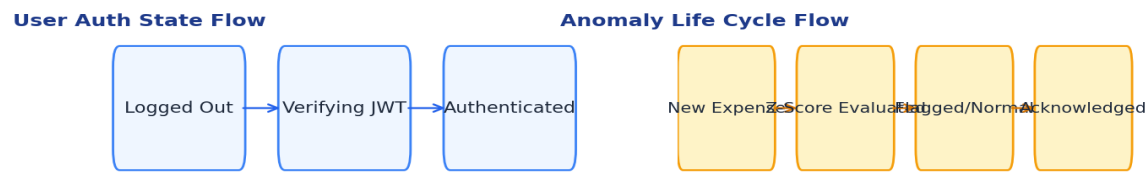


Figure 3.4 & 3.5: State Diagrams for Authentication and Anomaly Life Cycle

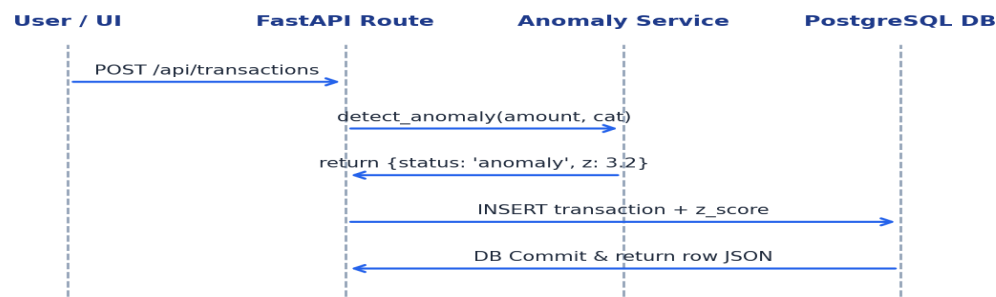


Figure 3.6: Sequence Diagram for User Login and Transaction Creation

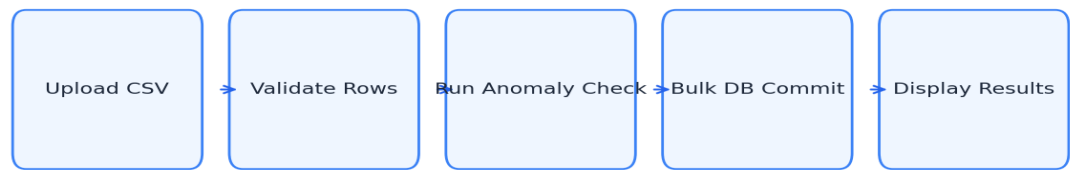


Figure 3.7: Activity Diagram for CSV Transaction Import

3.2 System Design

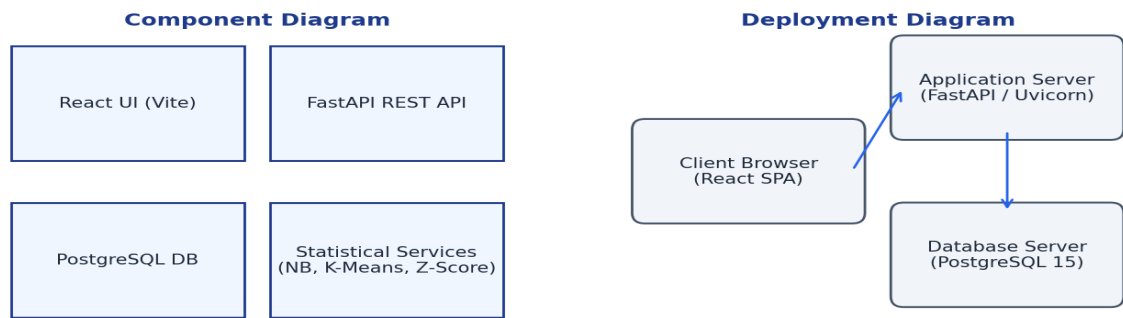


Figure 3.8 & 3.9: Component Diagram and Deployment Diagram of FinSight

3.3 Algorithm Details

3.3.1 Z-Score Anomaly Detection Algorithm

Flags expenses deviating significantly from a user's category mean. Formula: $Z = (x - \mu) / \sigma_{\text{floored}}$, where $\sigma_{\text{floored}} = \max(\sigma, |\mu| * 0.05)$ [Epsilon Floor]. Requires MIN_SAMPLES = 5. Severity thresholds: $|Z| \geq 2$ (moderate), $|Z| \geq 3$ (high).



Figure 3.10: Z-Score Anomaly Detection – Algorithm Flowchart

3.3.2 Naive Bayes Categorizer with Levenshtein Typo Matching

Suggests categories from description text. 1) Phrase Lookup Table (PHRASE_HINTS) 2) Levenshtein Edit Distance for typos 3) Multinomial Naive Bayes with Laplace smoothing calculated in log-space. Confidence threshold $\geq 28\%$.



Figure 3.11: Naive Bayes Categorizer – Algorithm Flowchart

3.3.3 1D K-Means Spending Clustering Algorithm

Groups expenses into micro, regular, major tiers. Initializes centroids at min, median, max. Iteratively updates centroids based on mean distance.



Figure 3.12: 1D K-Means Spending Clustering – Algorithm Flowchart

3.3.4 Budget Pace Prediction Algorithm

Extrapolates daily burn rate: $\text{Daily Rate} = \text{Spent} / \text{Current Day}$, $\text{Projected} = \text{Daily Rate} * \text{Days in Month}$. Status: on_track, at_risk, will_exceed, exceeded.

CHAPTER 4: IMPLEMENTATION AND TESTING

4.1 Implementation

4.1.1 Tools Used

Category	Technology	Purpose
Frontend	React 18 + Vite + Redux Toolkit + TailwindCSS	SPA, state management, styling
Backend	FastAPI (Python 3.11+)	REST API, business logic, algorithms
ORM	SQLAlchemy 2.0	Database abstraction layer
Database	PostgreSQL 15	Persistent relational storage
Auth	JWT + bcrypt	Secure token-based authentication
Charts	Recharts	Interactive spending visualization
Algorithms	Pure Python (statistics, math modules)	Z-Score, Naive Bayes, K-Means, Pace Prediction

Table 4.1: Development Tools and Technologies

4.2 Testing

4.2.1 Unit Testing

Test Case ID	Module Tested	Expected Result	Status
UT-01	Password Hashing (bcrypt)	Password correctly hashed & verified	PASS
UT-02	Z-Score ($N < 5$)	Returns status: 'not_enough_data'	PASS
UT-03	Z-Score (Epsilon Floor)	Prevents division by zero on identical amounts	PASS
UT-04	Naive Bayes Categorizer	Returns category with confidence score	PASS
UT-05	Levenshtein Distance	Matches 'pathao' to 'pathao' (distance 1)	PASS
UT-06	1D K-Means Clustering	Clusters amounts into 3 distinct centroids	PASS
UT-07	Budget Pace Prediction	Correctly projects end-of-month spend	PASS

Table 4.3: Unit Test Cases

CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATIONS

5.1 Conclusion

To conclude, the project FinSight is a successful implementation of a modern, statistically intelligent personal finance and budget tracking platform. All stated objectives – unified transaction management, Z-Score anomaly detection, Multinomial Naive Bayes categorizer with Levenshtein typo matching, 1D K-Means spending clustering, budget pace prediction, CSV import, and responsive UI – have been achieved. The system provides an end-to-end digital space for financial tracking, bridging the gap between simple spreadsheets and intelligent financial assistants.

5.2 Lesson learnt/Outcome

During development, we learned the importance of incremental delivery: core CRUD and authentication were validated early, after which complex analytical modules were layered on top. We gained deeper understanding of statistical and machine learning methods applied to real user data and proper edge-case handling (epsilon floor, minimum sample size).

5.3 Future Recommendations

1. Native mobile applications (iOS/Android) with offline support.
2. Direct bank API / Open Banking integration for automated transaction imports.
3. Shared family accounts with role-based access.

APPENDICES

Appendix A – API Endpoints Summary

POST /api/auth/register, POST /api/auth/login, GET /api/transactions, POST /api/transactions, GET /api/budgets, POST /api/budgets, GET /api/goals, POST /api/goals/{id}/contribute, GET /api/analytics/dashboard, GET /api/anomalies, POST /api/transactions/import, GET /api/categories/suggest.

Appendix B – Database Schema Notes

All primary keys are UUIDs. Foreign keys enforce referential integrity. Unique constraint on (user_id, category, month, year) for budgets. Anomaly results (z_score, severity, status) are stored directly on the transaction row for query performance.

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